Blessed G/Michael Catholic Secondary School

1st Semester Chemistry final examination for Grade 10th

Jan., 2018 E.C Time Allowed 1:30 min'

Name Bitanya Wondimagegnehu Section 10th B No 12

I. Write true it the statement is correct and false if the statement is wrong.

false 1. When balancing an equation, you can change the subscripts within a chemical formula to make atoms equal.

2. A substance that gets oxidized acts as the reducing oxidizing agent and reduced acts as reducing agent. false

3. A chemical reaction always produce new substance with different properties. True

4. The solubility of a solid in liquid decreases if the dissolution process is endothermic. false

5. When a solution is diluted by adding more solvent the number of mole of solute in solution is changed. false

6. The rate of dissolution of a solid solute generally increase when the surface area of solute is increased. True

7. Pressure has a significant effect on solubility of solid solute in liquid solvent. false

II. Match the words or phrases under column 'B' with the appropriate items under column 'A'

'A'	'B'
8. Forms a single product A	A. combination reaction
9. Breaking down single compounds in to two or more element C	B. redox reaction
10. Replacement reaction D	C. Decomposition reaction
11. Exchange of positive and negative ion E	D. Single displacement reaction
12. The transfer of electron between species B	E. double displacement reaction

III. Chose the best answer among the given alternative

B 13. A 0.5L and 0.1M HNO_3 solution is to be prepared by dilution process from a 4M HNO_3 How many ml concentrated HNO_3 and How many ml of water are required to prepare the dilution.

A. 6.5ml HNO_3 and 480 ml water
B. 125ml HNO_3 and 487.5 ml water
C. 10 ml HNO_3 and 400ml water
D. 8 ml HNO_3 and 480.5ml water

C 14. A solution of NH_4Cl made by dissolving 5.4g NH_4Cl and 36.01g H_2O has density of 1.027g/ml. what is the mole fraction NH_4Cl? Mass of Cl= 36, N= 14)

A. 0.05 B. 0.476 C. 0.0476 D. 0.051

B 15. Which of the following explains dipole – dipole force?

A. The intermolecular attraction force that exists between non- polar molecules
B. A type of intermolecular force that exists between polar molecules
C. An electrostatic attraction between metal ion and valance electron
D. An attractive for resulting from temporary dipoles induced in molecules

C 16. The partial pressure of acetylene. C_2H_2 in a container is 10 atm at 25°C. What is the solubility in molarity of C_2H_2, assume that C_2H_2 obeys Henery's Low? KH= 1.04mol/L atm

A. 0.104mol / L B. 9.62mol /L C. 10.4mol /L D. 104mol /L

C 17. What is the normality of a solution that contains 19.6 H_2SO_4 in 20ml of solution?

A. 0.01 B. 0.2 C. 20 D. 10

A 18. The difference between colloids and suspension is that.

A. Colloidal particle are small enough and do not settle down unlike in a suspension
B. Most colloids and suspensions appear cloudy
C. Both colloids and suspensions are mixture
D. Colloids are as transparent a solution

B 19. What is the molarity of a 3.4g of hydrogen peroxide (H_2O_2) in 200ml solution that is used to for hair bleaching? (Mass H_2O_2 =34

A. 0.05 M B. 0.5 M C. 0.15M D. 0.015M

D 20. How many moles of $AgNO_3$ are needed to react completely with 50 ml of 0.2M $MgCl_2$?

$2AgNO_3+MgCl_2 \longrightarrow 2AgCl+Mg(NO_3)_2$

A. 0.04mol B. 0.01mol C. 0.005mol D. 0.02 mole

A 21. What the volume of 0.5M H_2SO_4 is required to neutralized 25ml of 1M KOH?

A. 50ml B. 12.5ml C. 25ml D. 100ml

A 22. How many grams of $CaCO_3$ are produced from 500ml of 0.1M Na_2CO_3 reacting excess $CaCl_2$?

(Mas Ca= 40, Na=23, Cl=35.5, C= 12,O=16

$Na_2CO_3+CaCl_2 \longrightarrow CaCO_3+2NaCl$

A. 5g B. 2 g C. 50g D. 10g

B 23. When $BaCl_2(aq)$ and $Na_2SO_4(aq)$ are mixed a white precipitate forms what is the identity of this priciptate

A. NaCl B. $BaSO_4$ C. $BaCl_2$ D. $NaSO_4$

D 24. If 40g of sodium hydroxide is dissolved in 2000g of water, what is the molality of solution? (Mass NaOH = 40)

A. 1.0m B. 2.0m C. 0.25m D. 0.5m

A 25. What is the percent by mass of sugar that are required to prepare 200g sugar solution of a 10 g sugar?

A. 5 % B. 0.5 % C.0. 25 % D. 50%

B 26. Which of the following reaction is not balanced?

A. $2NaHCO_3 \longrightarrow Na_2CO_3+H_2O+CO_2$
B. $2C_4H_{10}+12O_2 \longrightarrow 8CO_2+10H_2O$
C. $2Al+6H_2O \longrightarrow 2Al(OH)_3+3H_2$
D. $4NH_3+5O_2 \longrightarrow 4NO+6H_2O$

B 27. Which of the following reaction is not correct

A. $Zn+CuSO_4 \longrightarrow ZnSO_4+Cu$
B. $2Ag+Cu(NO_3)_2 \longrightarrow AgNO_3+Cu$
C. $Fe+CuSO_4 \longrightarrow FeSO_4+Cu$
D. $Mg+2HCl \longrightarrow MgCl_2+H_2$

A 28. What the volume of 0.5M sugar contained $3.011 * 10^{22}$ molecule of sugar?

A. 0.01L B. 0.1L C. 0.5L D. 0.05L

A 29. For the following chemical question what is the spectator ion?

$Na2SO_4(aq)+Ba(NO_3)_2(aq) \longrightarrow BaSO_4(s)+2NaNO_3(aq)$

A. Na^+ and NO_3^- B. SO_4^{-2} and NO_3^- C. Na^+ and Ba^{2+} D. Ba^{2+} and SO_4^{-2}

D 30. Which one of the following molecules not form hydrogen bonding within its molecule

A. HF B. H_2O C. NH_3 D. PCl_5

C 31. All of the following compound is soluble in water except one:-

A. $AgNO_3$ B. CH_3COONa C. $CaCO_3$ D. K_3PO_4

A 32. Rate of dissolution is largely dependent up on

A. The inter- particle force
B. The surface area
C. The temperature of the system
D. The pressure of the system

B 33. To prepare 250ml of 10% (V/V) vinegar solution. How many milliliters of pure acetic acid (solute are needed?

A. 2.5ml B. 25ml C. 250ml D. 100ml

IV. Short answer

34. Letrgenous is a mixture that consists of physically distinct parts, each within different properties.
35. super saturated solution is a solution usually create increasing temperature and pressure.
36. recrystalization is the process dissolved solute returning to the solid state.
37. A aque solution of sodium hydroxide is 20% by mass the density of solution is 1.01 g/ml
    calculate
    A. Molarity ~~0.005M~~ 0,005M
    B. Molality 0,006
    C. Mole fraction NaOH ~~0.1~~ 0.1
    D. Normality 0,005 N

36/40

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1. false	8. A	15. B	22. A	29. A
2. false	9. C	16. C	23. B	30. D
3. TRUE	10. D	17. C	24. D	31. C
4. false	11. E	18. A	25. A	32. A
5. false	12. B	19. B	26. B	33. B
6. TRUE	13. B	20. D	27. B	
7. false	14. C	21. A	28. ~~B~~	

Short answer

34, hetrogenouse
35, Super Saturated
36, recrystalization
37, A, ~~0.005 M~~ 5M
B, ~~0.006 m~~ 6.25m
C, 0.1
D, ~~0.005 M~~ 5